

SPATBench: Methodological Choices Can Outweigh Algorithmic Choices in ICI Transcriptomic Prediction

Circularity Detection Score (CDS)

Input: expression matrix
+ response labels
+ DECLARE endpoint genes

C1 MI(genes, label) 30%
C2 |Spearman (genes, label)| 30%
C3 top-500 / random MI 40%

$$\text{CDS} = 0.3 \cdot \text{C1n} + 0.3 \cdot \text{C2} + 0.4 \cdot \text{C3}$$

**null-referenced:
vs cohort-matched label shuffles**

1 Endpoint switch collapses

Cytolytic surrogate vs RECIST v1.1:

EN(Var) 0.985 → 0.465

GEP 0.968 → 0.503

IMPRES (no overlap): unchanged

Effect = 3.5x method gap

2 Feature selection rewrites

Var- vs MI-based selection:

27% gene overlap

yet indistinguishable AUROC

→ the narrative changes,

not the ranking

3 CDS on real cohorts

Circular cytolytic:

CDS 0.954 >> null 0.842

Clinical: inside nulls

Labels non-specific

(60/60 pseudo-endpoints HIGH)

High-resolution validation: ElasticNet (MI) on Gide 2019 (n = 73) becomes FDR-significant

at 5,000 full-pipeline shuffles

p = 0.0232, within-cohort BH q = 0.0278 | no trainable method survives FDR on any clinical endpoint at n ≤ 43